

Removing Trace Preservation from a Universal Spectral Bound for 2-Positive Maps*

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Abstract

Vom Ende, Chruściński, Kimura and Muratore-Ginanneschi proved that every 2-positive trace-preserving map Φ on $M_d = M_d(\mathbb{C})$ satisfies $\text{Tr } \Phi \leq d m(\Phi) + d^2 - d$, where $m(\Phi)$ is the smallest real part of an eigenvalue of Φ , and deduced the universal relaxation-rate bound $\text{Tr } L \leq d m(L)$ for generators of 2-positive trace-preserving semigroups. They asked whether trace preservation is needed, and conjectured that every conditionally 2-positive map L , i.e. every L for which e^{tL} is 2-positive for all $t \geq 0$, satisfies

$$\text{Tr } L \leq d m(L) + (d^2 - d) s(L),$$

where $s(L)$ is the largest real part of an eigenvalue of L . We prove this conjecture. The argument normalizes a 2-positive map by a positive definite eigenmatrix of its Hilbert–Schmidt adjoint, which turns it into a 2-positive trace-preserving map in the manner of a quantum Doob transform. The existence of such an eigenmatrix is guaranteed after adding an arbitrarily small multiple of the unnormalized depolarizing map, and the inequality then passes to the limit by continuity of the spectrum. No irreducibility, diagonalizability or faithful-invariant-state assumption is required. We also show that the constant d is optimal within the class of non-trace-preserving completely positive semigroups, and we recast the result as a universal constraint on decay rates measured relative to the dominant spectral growth rate.

Keywords: 2-positive maps; conditionally completely positive maps; quantum dynamical semigroups; relaxation rates; eigenvalue inequalities; quantum Doob transform.

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1 Introduction

The eigenvalues of the generator of a Markovian quantum evolution determine its relaxation rates, which are the experimentally accessible rates of decoherence, dissipation and thermalization [2, 1, 12]. Complete positivity constrains these rates in a nontrivial way. For a generator L of a completely positive trace-preserving (CPTP) semigroup on a d -level system, the relaxation rates $\Gamma_j = -\text{Re } \lambda_j$ obey the universal bound

$$\Gamma_k \leq \frac{1}{d} \sum_j \Gamma_j \quad \text{for every } k. \quad (1.1)$$

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This bound was conjectured in [5, 4], building on the qubit case of [10], and proved in [11] using Lyapunov-theoretic methods.

Vom Ende, Chruściński, Kimura and Muratore-Ginanneschi [13] later gave an entirely algebraic proof that applies under the weaker hypothesis of 2-positivity. The key step in their argument is an inequality for maps rather than generators (Theorem A below): every 2-positive trace-preserving map Φ on M_d satisfies

$$\mathrm{Tr} \Phi \leq d m(\Phi) + d^2 - d. \quad (1.2)$$

Here $m(\Phi)$ is the smallest real part of an eigenvalue of Φ . Inequality (1.1) then follows by differentiation along the semigroup; see also [6] for a direct proof at the generator level.

In the outlook of [13], the authors note that trace preservation enters the inequality mainly by fixing the dominant eigenvalue. That eigenvalue equals 1 for trace-preserving maps and 0 for their generators. They ask whether this hypothesis can be dropped once the dominant eigenvalue is reinstated explicitly, and they formulate the following conjecture. Writing $s(\mathcal{A})$ for the largest real part of an eigenvalue of $\mathcal{A}$, and calling a linear map L on M_d *conditionally 2-positive* if e^{tL} is 2-positive for all $t \geq 0$, the conjecture reads as follows.

Conjecture ([13, Sec. 3, Eq. (10)]). *Every conditionally 2-positive map $L: M_d \rightarrow M_d$ satisfies*

$$\mathrm{Tr} L \leq d m(L) + (d^2 - d) s(L).$$

The class of conditionally 2-positive maps contains every 2-positive map, every completely positive map, and every conditionally completely positive map in the sense of Evans [8, 15]. The authors of [13] report numerical evidence in the completely positive case. They also note that a proof would likely require a substantially different approach, because their argument relies on the existence of a fixed point, which is no longer available without trace preservation.

Main result. We prove the conjecture.

Theorem 1. *Let $L: M_d \rightarrow M_d$ be conditionally 2-positive. Then*

$$\mathrm{Tr} L \leq d m(L) + (d^2 - d) s(L). \quad (1.3)$$

Moreover, for each $d \geq 2$, equality holds in (1.3) for a family of generators of completely positive semigroups that are not trace-preserving. In particular, in an inequality of the form $\mathrm{Tr} L \leq a m(L) + (d^2 - a) s(L)$, the coefficient $a = d$ cannot be replaced by any larger number.

The proof proceeds through the corresponding statement for maps.

Theorem 2. *Every 2-positive map $\Phi: M_d \rightarrow M_d$ satisfies*

$$\mathrm{Tr} \Phi \leq d m(\Phi) + (d^2 - d) s(\Phi). \quad (1.4)$$

The difficulty identified in [13] is real. A general 2-positive map has no distinguished fixed point, its dominant eigenvalue need not be 1, and it may be reducible or nilpotent. Our approach is to reduce to Theorem A instead of reproving it. If the Hilbert–Schmidt adjoint Φ^* has a *positive definite* eigenmatrix H with eigenvalue $r > 0$, then conjugation by $H^{1/2}$ followed by division by r produces a 2-positive *trace-preserving* map with the same spectrum up to the factor r . This normalization is a version of the quantum Doob transform used in the large-deviation theory of open quantum systems [3]. A positive definite eigenmatrix need not exist, but it does exist after adding $\varepsilon \mathrm{tr}(\cdot) I$ to Φ , by Brouwer’s fixed-point theorem. The inequality is stable as $\varepsilon \downarrow 0$ because it involves only the trace and the extreme real parts of

the spectrum, all of which depend continuously on the map. In particular, the normalizing matrices H_ε do not need to converge, so degenerate limits cause no difficulty.

Theorem 1 has a natural interpretation in terms of decay rates. Set $\Gamma_j = s(L) - \operatorname{Re} \lambda_j \geq 0$, the decay rates measured relative to the dominant growth rate. Then (1.3) is equivalent to $\max_k \Gamma_k \leq \frac{1}{d} \sum_j \Gamma_j$ (Corollary 11), which extends (1.1) verbatim to non-trace-preserving dynamics. Such dynamics arise, for instance, as tilted generators in the study of quantum-trajectory fluctuations [3].

Organization. Section 2 fixes notation and records the input from [13]. Section 3 contains a standard spectral lemma for positive trace-preserving maps. Section 4 proves Theorem 2 in the presence of a positive definite eigenmatrix, and Section 5 removes that assumption. Section 6 deduces Theorem 1. Section 7 establishes sharpness, and Section 8 discusses the decay-rate formulation.

2 Notation and preliminaries

Throughout, d is a positive integer and all maps are complex-linear. Let $M_d = M_d(\mathbb{C})$ with the Hilbert–Schmidt inner product $\langle A, B \rangle = \operatorname{tr}(A^\dagger B)$. For a linear map $\mathcal{A}: M_d \rightarrow M_d$, $\mathcal{A}^*$ denotes its Hilbert–Schmidt adjoint. We write tr for the trace of a $d \times d$ matrix and $\operatorname{Tr} \mathcal{A}$ for the trace of $\mathcal{A}$ as an operator on the d^2 -dimensional space M_d . Thus $\operatorname{Tr} \mathcal{A} = \sum_\lambda \lambda$, with eigenvalues counted according to algebraic multiplicity. The spectrum of $\mathcal{A}$ is $\sigma(\mathcal{A})$, and we set

$$m(\mathcal{A}) = \min_{\lambda \in \sigma(\mathcal{A})} \operatorname{Re} \lambda, \quad s(\mathcal{A}) = \max_{\lambda \in \sigma(\mathcal{A})} \operatorname{Re} \lambda.$$

For matrices, $X \geq 0$ means positive semidefinite and $X > 0$ means positive definite. The operator norm is $\|\cdot\|_\infty$.

Definition 3. A linear map $\Phi: M_d \rightarrow M_d$ is *positive* if $\Phi(X) \geq 0$ whenever $X \geq 0$. It is *k-positive* if $\operatorname{id}_{M_k} \otimes \Phi$ is positive on $M_k \otimes M_d$, and *completely positive* if it is *k-positive* for every k . It is *trace-preserving* if $\operatorname{tr} \Phi(X) = \operatorname{tr} X$ for all X . A linear map $L: M_d \rightarrow M_d$ is *conditionally 2-positive* if e^{tL} is 2-positive for every $t \geq 0$.

We record some elementary facts that will be used repeatedly.

- (P1) *Hermiticity and real traces.* A positive map is Hermitian-preserving, since every Hermitian X is a difference of two positive semidefinite matrices. If e^{tL} is Hermitian-preserving for all $t \geq 0$, then so is $L = \lim_{t \downarrow 0} t^{-1}(e^{tL} - \operatorname{id})$. A Hermitian-preserving map is represented by a real matrix in any basis of Hermitian matrices. Hence its characteristic polynomial has real coefficients, its spectrum is invariant under complex conjugation, and its trace is real. In particular, all quantities appearing in (1.3) and (1.4) are real.
- (P2) *Positivity of the adjoint.* If Φ is positive, then so is Φ^* . Indeed, for $Y \geq 0$ and $v \in \mathbb{C}^d$ we have $\langle v, \Phi^*(Y)v \rangle = \operatorname{tr}(Y \Phi(vv^\dagger)) \geq 0$, because the trace of a product of two positive semidefinite matrices is nonnegative.
- (P3) *Congruences.* For any $S \in M_d$, the map $X \mapsto SXS^\dagger$ is completely positive, since its k -fold amplification is congruence by $I_k \otimes S$. The 2-positive maps form a convex cone that is closed under composition and topologically closed.
- (P4) *Continuity of the spectrum.* The eigenvalues of a matrix, counted with algebraic multiplicity, depend continuously on its entries as an unordered d^2 -tuple, because the

roots of a monic polynomial depend continuously on its coefficients [9, Ch. II, Thm. 5.14]. Consequently, $\mathcal{A} \mapsto m(\mathcal{A})$, $\mathcal{A} \mapsto s(\mathcal{A})$ and $\mathcal{A} \mapsto \text{Tr } \mathcal{A}$ are continuous.

The only specialized result we use is the main theorem of [13].

Theorem A (vom Ende, Chruściński, Kimura, Muratore-Ginanneschi [13]) ([13, Thm. 1]). *If $\mathcal{T}: M_d \rightarrow M_d$ is 2-positive and trace-preserving, then*

$$\text{Tr } \mathcal{T} \leq d m(\mathcal{T}) + d^2 - d. \quad (2.1)$$

3 A spectral lemma for positive trace-preserving maps

The following fact is well known; see, e.g., [14, Prop. 4.26]. We include a short proof for completeness.

Lemma 4. *Let $\mathcal{T}: M_d \rightarrow M_d$ be positive and trace-preserving. Then $1 \in \sigma(\mathcal{T})$ and $|\lambda| \leq 1$ for every $\lambda \in \sigma(\mathcal{T})$. In particular, $s(\mathcal{T}) = 1$.*

Proof. Let $\mathcal{U} = \mathcal{T}^*$. By (P2), $\mathcal{U}$ is positive. Trace preservation of $\mathcal{T}$ is equivalent to $\mathcal{U}(I) = I$, so $1 \in \sigma(\mathcal{U})$. Since $\sigma(\mathcal{T}) = \sigma(\mathcal{U})$, it follows that $1 \in \sigma(\mathcal{T})$.

For Hermitian A we have $-\|A\|_\infty I \leq A \leq \|A\|_\infty I$. Since each power $\mathcal{U}^n$ is positive and unital, this gives $\|\mathcal{U}^n(A)\|_\infty \leq \|A\|_\infty$. Now write an arbitrary $X \in M_d$ as $X = A + iB$ with $A = \frac{1}{2}(X + X^\dagger)$ and $B = \frac{1}{2i}(X - X^\dagger)$ Hermitian. Then $\|A\|_\infty, \|B\|_\infty \leq \|X\|_\infty$, and therefore $\|\mathcal{U}^n(X)\|_\infty \leq 2\|X\|_\infty$. If $\mathcal{U}(X) = \lambda X$ with $X \neq 0$, it follows that $|\lambda|^n \leq 2$ for all $n \in \mathbb{N}$, so $|\lambda| \leq 1$. Passing to complex conjugates proves the claim for $\mathcal{T}$. $\square$

4 Normalization by a positive definite eigenmatrix

Lemma 5. *Let $\Phi: M_d \rightarrow M_d$ be 2-positive, and suppose that there are a positive definite $H \in M_d$ and a real number $r > 0$ such that*

$$\Phi^*(H) = rH. \quad (4.1)$$

Then $s(\Phi) = r$, and Φ satisfies (1.4).

Proof. Let $\mathcal{C}_H(X) = H^{1/2}XH^{1/2}$. This map is invertible, and its inverse is $\mathcal{C}_H^{-1}(X) = H^{-1/2}XH^{-1/2}$. Define

$$\mathcal{T} = \frac{1}{r} \mathcal{C}_H \circ \Phi \circ \mathcal{C}_H^{-1}. \quad (4.2)$$

By (P3), $\mathcal{C}_H$ and $\mathcal{C}_H^{-1}$ are completely positive. Since $r > 0$, $\mathcal{T}$ is 2-positive. It is also trace-preserving: for every $X \in M_d$, using cyclicity of the trace, the Hermiticity of H and (4.1),

$$\begin{aligned} \text{tr } \mathcal{T}(X) &= \frac{1}{r} \text{tr} \left[H \Phi(H^{-1/2}XH^{-1/2}) \right] = \frac{1}{r} \text{tr} \left[\Phi^*(H) H^{-1/2}XH^{-1/2} \right] \\ &= \text{tr} \left[H H^{-1/2}XH^{-1/2} \right] = \text{tr } X. \end{aligned}$$

Because (4.2) is a similarity transformation followed by multiplication by r^{-1} , we have

$$\sigma(\mathcal{T}) = r^{-1}\sigma(\Phi), \quad \text{Tr } \mathcal{T} = r^{-1} \text{Tr } \Phi, \quad m(\mathcal{T}) = r^{-1}m(\Phi), \quad (4.3)$$

with algebraic multiplicities preserved. By Lemma 4, $1 \in \sigma(\mathcal{T})$ and every eigenvalue of $\mathcal{T}$ has modulus at most 1. Hence $r \in \sigma(\Phi)$ and every eigenvalue of Φ has modulus at most r , so $s(\Phi) = r$. Applying Theorem A to $\mathcal{T}$ and using (4.3),

$$\frac{\text{Tr } \Phi}{r} \leq d \frac{m(\Phi)}{r} + d^2 - d.$$

Multiplying by $r = s(\Phi) > 0$ gives (1.4). $\square$

Remark 6. The map (4.2) is a map-level analogue of the quantum Doob transform of [3], which there normalizes tilted generators using the dominant eigenmatrix. In [13] a related ω -weighted inner product is used to symmetrize trace-preserving maps that have a faithful fixed point. Here we use the normalization in the opposite direction, to *produce* trace preservation.

5 Removing the eigenmatrix hypothesis: proof of Theorem 2

A general 2-positive map need not admit a positive definite eigenmatrix for its adjoint. For example, the adjoint may be nilpotent, or every positive semidefinite eigenmatrix may be singular because of reducibility. We remove this hypothesis by perturbation.

Proof of Theorem 2. Let Φ be 2-positive and let

$$\Delta(X) = \text{tr}(X) I = \sum_{i,j=1}^d E_{ij} X E_{ij}^\dagger,$$

where $E_{ij} = |i\rangle\langle j|$ are the matrix units. By (P3), Δ is completely positive, and it is self-adjoint with respect to the Hilbert–Schmidt inner product. For $\varepsilon > 0$ put

$$\Phi_\varepsilon = \Phi + \varepsilon \Delta,$$

which is 2-positive by (P3).

Let $\mathcal{D}_d = \{H \in M_d : H \geq 0, \text{tr } H = 1\}$ be the set of density matrices. It is compact, convex and nonempty. For $H \in \mathcal{D}_d$, positivity of Φ^* (by (P2)) gives

$$\Phi_\varepsilon^*(H) = \Phi^*(H) + \varepsilon I \geq \varepsilon I. \quad (5.1)$$

In particular, $\text{tr } \Phi_\varepsilon^*(H) \geq \varepsilon d > 0$, so

$$F_\varepsilon(H) = \frac{\Phi_\varepsilon^*(H)}{\text{tr } \Phi_\varepsilon^*(H)}$$

is a continuous self-map of $\mathcal{D}_d$. By Brouwer’s fixed-point theorem there is $H_\varepsilon \in \mathcal{D}_d$ with $F_\varepsilon(H_\varepsilon) = H_\varepsilon$. Setting $r_\varepsilon = \text{tr } \Phi_\varepsilon^*(H_\varepsilon) > 0$, we obtain

$$\Phi_\varepsilon^*(H_\varepsilon) = r_\varepsilon H_\varepsilon,$$

and (5.1) yields $H_\varepsilon \geq (\varepsilon/r_\varepsilon) I > 0$. Therefore Lemma 5 applies to Φ_ε , and

$$\text{Tr } \Phi_\varepsilon \leq d m(\Phi_\varepsilon) + (d^2 - d) s(\Phi_\varepsilon) \quad \text{for every } \varepsilon > 0. \quad (5.2)$$

As $\varepsilon \downarrow 0$, $\Phi_\varepsilon \rightarrow \Phi$. By (P4), $\text{Tr } \Phi_\varepsilon \rightarrow \text{Tr } \Phi$, $m(\Phi_\varepsilon) \rightarrow m(\Phi)$ and $s(\Phi_\varepsilon) \rightarrow s(\Phi)$. Letting $\varepsilon \downarrow 0$ in (5.2) proves (1.4). $\square$

Remark 7. The limiting argument uses only the perturbed maps Φ_ε and their spectra. It does not require convergence of the eigenmatrices H_ε , their inverses, or the normalized maps $\mathcal{T}_\varepsilon$, nor an a priori positive lower bound on r_ε . Singular limiting eigenmatrices, a vanishing spectral radius, and nontrivial Jordan blocks therefore cause no difficulty. As a by-product, $s(\Phi) = \lim_\varepsilon r_\varepsilon$ is the spectral radius of Φ and is an eigenvalue, which recovers the Perron–Frobenius property of positive maps (cf. [7]). The argument as a whole uses 2-positivity only through Theorem A.

6 Generators: proof of Theorem 1

Proof of inequality (1.3). Let L be conditionally 2-positive, and let $\lambda_1, \dots, \lambda_{d^2}$ be its eigenvalues counted with algebraic multiplicity. For each $t > 0$ the map e^{tL} is 2-positive, so Theorem 2 gives

$$\mathrm{Tr} e^{tL} \leq d m(e^{tL}) + (d^2 - d) s(e^{tL}). \quad (6.1)$$

By the spectral mapping theorem (which holds via the Jordan form, without any diagonalizability assumption), the eigenvalues of e^{tL} are $e^{t\lambda_1}, \dots, e^{t\lambda_{d^2}}$, with multiplicities. Since there are finitely many λ_j , Taylor's theorem gives a constant C such that

$$|\mathrm{Re} e^{t\lambda_j} - 1 - t \mathrm{Re} \lambda_j| \leq C t^2 \quad \text{for all } j \text{ and all } t \in (0, 1].$$

Taking a minimum or maximum over j preserves this uniform error, so

$$m(e^{tL}) = 1 + t m(L) + O(t^2), \quad s(e^{tL}) = 1 + t s(L) + O(t^2),$$

and furthermore $\mathrm{Tr} e^{tL} = \sum_j e^{t\lambda_j} = d^2 + t \mathrm{Tr} L + O(t^2)$. Substituting these expansions into (6.1), the constant terms d^2 cancel on both sides. Dividing by $t > 0$ and letting $t \downarrow 0$ yields (1.3). $\square$

Remark 8. Every 2-positive map Φ is conditionally 2-positive. Indeed, each power Φ^n is 2-positive, and $e^{t\Phi} = \sum_{n \geq 0} \frac{t^n}{n!} \Phi^n$ is a norm limit of nonnegative combinations of 2-positive maps. Hence Theorem 1 contains Theorem 2 as well as the generator statement, which settles both readings of the conjecture in [13]. If e^{tL} is trace-preserving for every $t \geq 0$, then $L^*(I) = 0$ and Lemma 4 gives $|e^{t\lambda}| \leq 1$ for each eigenvalue λ of L . Thus $0 \in \sigma(L)$, $\mathrm{Re} \lambda \leq 0$, and $s(L) = 0$. If Φ is positive and trace-preserving, then $s(\Phi) = 1$. In these respective cases, Theorems 1 and 2 reduce to [13, Thms. 2 and 1].

7 Sharpness for non-trace-preserving dynamics

The bound (1.3) is attained by generators of completely positive semigroups that are *not* trace-preserving, so the extension is sharp precisely in the regime it was designed for.

Proposition 9. *Fix $d \geq 2$ and an orthonormal basis $\{|0\rangle, \dots, |d-1\rangle\}$ of $\mathbb{C}^d$. Put $V = |0\rangle\langle 1|$ and $P = |1\rangle\langle 1|$, and define*

$$G(X) = V X V^\dagger - \frac{1}{2}(P X + X P), \quad L_c = G + c \mathrm{id}_{M_d} \quad (c \in \mathbb{R}). \quad (7.1)$$

Then e^{tL_c} is completely positive for all $t \geq 0$, and it is trace-preserving for some $t > 0$ only if $c = 0$. Moreover,

$$\mathrm{Tr} L_c = c d^2 - d, \quad m(L_c) = c - 1, \quad s(L_c) = c,$$

so that equality holds in (1.3) for every $c \in \mathbb{R}$.

Proof. On matrix units $E_{ij} = |i\rangle\langle j|$ we have

$$G(E_{ij}) = \delta_{i1}\delta_{j1} E_{00} - \frac{1}{2}(\delta_{i1} + \delta_{j1}) E_{ij}.$$

This yields the eigen-decomposition in Table 1. The listed eigenvectors form a basis of M_d , so G is diagonalizable and

$$\mathrm{Tr} G = -\frac{1}{2} \cdot 2(d-1) - 1 = -d, \quad m(G) = -1, \quad s(G) = 0.$$

Eigenvalue of G	Multiplicity	Eigenvectors
0	$(d-1)^2$	$E_{ij}, \quad i, j \neq 1$
$-\frac{1}{2}$	$2(d-1)$	$E_{1j}, E_{j1}, \quad j \neq 1$
-1	1	$E_{11} - E_{00}$

Table 1: Spectrum of the amplitude-damping generator G of (7.1).

A direct check on matrix units shows that

$$e^{tG}(X) = K_0(t)XK_0(t)^\dagger + K_1(t)XK_1(t)^\dagger, \quad K_0(t) = I - P + e^{-t/2}P, \quad K_1(t) = \sqrt{1 - e^{-t}}V.$$

For example, $e^{tG}(E_{11}) = E_{00} + e^{-t}(E_{11} - E_{00}) = e^{-t}E_{11} + (1 - e^{-t})E_{00}$, which agrees with the Kraus form. Since $K_0^\dagger K_0 + K_1^\dagger K_1 = (I - P + e^{-t}P) + (1 - e^{-t})P = I$, the map e^{tG} is CPTP. Because id commutes with G , we get $e^{tL_c} = e^{ct}e^{tG}$. This map is completely positive and multiplies traces by e^{ct} , which equals 1 for some $t > 0$ only if $c = 0$. The spectral data of L_c are those of G shifted by c , which gives the stated values. Finally,

$$d m(L_c) + (d^2 - d) s(L_c) = d(c - 1) + (d^2 - d)c = cd^2 - d = \text{Tr } L_c. \quad \square$$

Corollary 10. *Let $a > d$. For every $d \geq 2$ there is a generator L of a non-trace-preserving completely positive semigroup such that $\text{Tr } L > a m(L) + (d^2 - a) s(L)$.*

Proof. For $L = L_c$ with $c \neq 0$, we have $m(L) - s(L) = -1$, and therefore $a m(L) + (d^2 - a)s(L) = d^2 s(L) - a < d^2 s(L) - d = \text{Tr } L$. $\square$

8 Decay rates relative to the dominant growth rate

For non-trace-preserving generators, such as the tilted generators used to study the full counting statistics of quantum jump trajectories [3], the leading eigenvalue is generally nonzero. It is then natural to measure decay relative to it.

Corollary 11. *Let L be conditionally 2-positive with eigenvalues $\lambda_1, \dots, \lambda_{d^2}$, counted with algebraic multiplicity, and define the relative decay rates $\Gamma_j = s(L) - \text{Re } \lambda_j \geq 0$. Then*

$$\max_k \Gamma_k \leq \frac{1}{d} \sum_{j=1}^{d^2} \Gamma_j, \quad (8.1)$$

and (8.1) is equivalent to (1.3).

Proof. By (P1), $\text{Tr } L = \sum_j \text{Re } \lambda_j$. Hence $\sum_j \Gamma_j = d^2 s(L) - \text{Tr } L$ and $\max_k \Gamma_k = s(L) - m(L)$. Inequality (8.1) therefore reads $d s(L) - d m(L) \leq d^2 s(L) - \text{Tr } L$, which is a rearrangement of (1.3). $\square$

For trace-preserving semigroups, $s(L) = 0$ and (8.1) reduces to the universal constraint (1.1). Thus the restriction on relaxation rates imposed by 2-positivity remains valid without trace preservation. The relative rates are invariant under scalar shifts $L \mapsto L + c \text{id}$ with $c \in \mathbb{R}$, since both $s(L)$ and every $\text{Re } \lambda_j$ shift by c . General non-trace-preserving perturbations need not be scalar shifts and may change the relative rates; the theorem asserts that the same universal inequality still holds. By Proposition 9, (8.1) is attained by non-trace-preserving dynamics.

9 Concluding remarks

We have shown that the trace-preservation hypothesis in the universal spectral bound of [13] can be removed, both for 2-positive maps and for their generators, once the dominant eigenvalue is restored as s . The proof treats the trace-preserving theorem as a black box and combines it with two soft tools: a Doob-type normalization by a positive definite eigenmatrix of the adjoint, and a perturbation that guarantees such an eigenmatrix exists. The same reduction applies to a continuous, positively homogeneous spectral inequality on an ambient class of positive maps, provided that class is stable under the normalization congruences, positive rescaling, and the perturbation $\Phi \mapsto \Phi + \varepsilon\Delta$, and the inequality is known for its trace-preserving members. These closure requirements concern the ambient class, not its trace-preserving subset.

The other questions raised in [13] concern whether complete positivity can be spectrally distinguished from 2-positivity, and whether positive trace-preserving maps in dimension $d \geq 3$ can violate (1.2). We do not address these questions here. Since normalization and perturbation use only positivity, validity of the latter bound for all positive trace-preserving maps would imply its homogeneous extension to all positive maps; conversely, a trace-preserving counterexample would already be a counterexample in the larger class.

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In accordance with arXiv policy, AI tools are not listed as authors. OpenAI ChatGPT was used to assist in preparing this paper. This assistance included exploring proof strategies, drafting and checking derivations, and editing the exposition. All mathematical content was reviewed by the author, who takes full responsibility for the correctness and originality of the work. To the author's knowledge, the conjecture of [13, Sec. 3] had not been resolved in the literature at the time of writing. The present argument has not yet undergone independent peer review.

Data and Code Availability

The supporting code package used to verify the analytical claims in this paper is open-source and publicly available on GitHub. It has been permanently archived on Zenodo at doi:10.5281/zenodo.23226133.

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